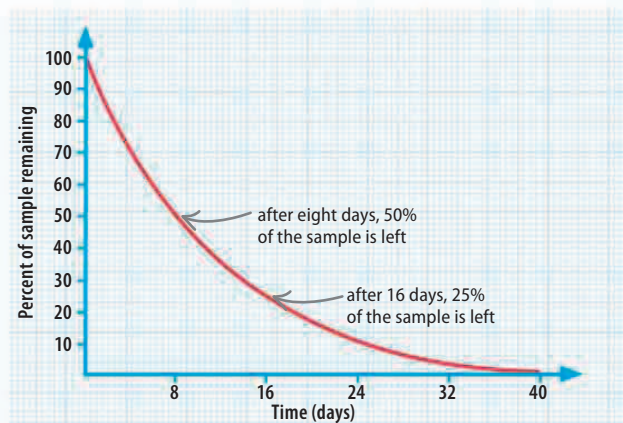


## Half-life

Radioactive isotopes decay at a fixed rate that is measured as a half-life. This is the amount of time it takes for a sample to reduce its mass by half as it decays into other elements. Every radioactive isotope has a specific half-life. The more radioactive an isotope is, the shorter its half-life.

### ▷ Fixed decay rate

The half-life of a radioactive substance is the same whether there is a lot of it or a little. Here, the half-life is eight days. After eight days, 50 percent of the original sample is left, after another eight days only 25 percent of the sample is left, and so on.



## Decay series

Radioactive isotopes often decay into daughter atoms that are also radioactive. The decay continues through a series of radioactive isotopes until finally stable atoms are formed. Members of a decay series may exist for just a fraction of a second, while others are around for years, only gradually breaking down into the next element in line.

### ▷ Uranium 238 series

This is the most common isotope of uranium. It decays in a series of alpha and beta emissions producing a series of radioactive isotopes before reaching a stable form of lead (Pb, mass number: 206). Each isotope decays at a different rate so each has a different half-life.

